



How to Design Plan – Act – Observe – Adjust Architectures for Agentic AI

AI is connecting siloed processes and transforming the scope of tomorrow's economy—one innovation at a time. According to PwC, AI could add \$15.7 trillion to the global economy by 2030. Yet many organizations still struggle to turn AI investments into meaningful, enterprise-grade outcomes. The reason? An overreliance on outdated, generic AI models.

Most current systems rely on stateless models—great for fast responses, but inadequate in dynamic, real-world environments. Enterprises need intelligent systems that can plan, act, observe, and adjust autonomously. The answer is implementing [Agentic AI](#) and [custom architecture](#) to meet your specific requirements.

What is Agentic AI Architecture?

Agentic AI is a system of [AI agents](#) that make autonomous decisions through planning, acting, observing results, and adjusting their behavior. These systems use memory, tools, and environmental awareness to operate over multiple steps.

Components of Agentic AI Architecture

and self-correction.

- **Memory Systems** are short- and long-term memory stores. They enable agents to retain context, learn across sessions, and avoid repetitive or inconsistent outputs.
- **Tool Orchestration and Decision Logic Layers** manage how agents select and execute tools or APIs to perform complex tasks.
- **Governance and Observability Modules** [provide oversight](#), ensure compliance, traceability, and support alignment with enterprise policies.
- **Data Flow and Integration Principles** define how data moves through the system, ensuring consistency, scalability, and secure handling across components.

This architecture allows enterprises to do everything from being sustainable and compliant to implementing multi-step reasoning, transparent decision-making, and real-time adaptability.

Why Custom Agentic AI Architecture is Essential for Enterprise Decision-Making

With custom [Agentic AI](#) architecture, enterprises can align AI systems with domain-specific data sources, business logic, and operational workflows. Unlike static, rule-based systems that are siloed and costly to update, Agentic AI architectures are scalable and fully integrated across operations.

Three Core Pillars of Agentic AI: Why Enterprises are implementing GenAI

Pillar 1: Goal-Oriented Autonomy

Agentic AI systems are capable of interpreting a broad, high-level objective provided by a user or enterprise, breaking it down into manageable, actionable sub-tasks. They autonomously design a detailed plan and execute it with minimal to no human intervention, significantly reducing the need for ongoing manual oversight.

The Key Benefit for Enterprises: Operational Scalability & Efficiency

operations seamlessly, without requiring a proportional increase in human resources or operational expenses.

Pillar 2: Adaptive Learning & Self-Correction

These intelligent systems continuously ingest and analyze new data streams, enabling them to detect when initial plans are not yielding desired results. They then self-correct by adjusting their strategies and workflows, refining their approach dynamically in real time to enhance performance and ensure better outcomes.

The Key Benefit for Enterprises: Continuous Optimization

Over time, these AI-driven processes continuously improve as the system learns from past experiences and outcomes. This ongoing optimization ensures that business operations remain both efficient and effective, even as market conditions and organizational needs evolve.

Pillar 3: Proactive Contextual Reasoning

Unlike traditional rule-based automation, Agentic AI anticipates future needs and potential risks by leveraging a deep understanding of context. It proactively initiates appropriate actions based on this insight, enabling more nuanced and adaptive

responses to dynamic environments.

The Key Benefit for Enterprises: Enhanced Decision-Making

By integrating advanced perception capabilities, sophisticated reasoning algorithms, and autonomous action-taking, Agentic AI significantly empowers organizations. This integration facilitates faster, smarter, and more resilient decision-making processes that drive improved business performance.

Case Study: FinRobot, Custom Agentic AI for Enterprise Resource Planning

A recent [research study](#) developed a research prototype called FinRobot. It is an agent-based framework designed to explore how Agentic AI concepts can improve enterprise resource planning (ERP) in finance. This was a controlled, academic experiment designed to validate important ideas surrounding the Agentic AI discussion.

The Challenge:

The research highlights that traditional ERP systems typically rely on rigid, rule-based workflows, which struggle to

- Integrate real-time data
- Handle unstructured inputs and
- Adapt to evolving business and regulatory requirements.

Agentic AI Approach:

In this study, FinRobot is designed around a manager agent that interprets high-level tasks and coordinates multiple specialized sub-agents. Each of these agents is responsible for specific functions, like wire transfer processing or employee reimbursements.

These sub-agents operate cross-functionally, adjust their actions based on real-time context, and incorporate generative AI for advanced reasoning and decision-making.

Outcomes of the Study:

- The framework showed a 40% reduction in simulated end-to-end process times.
- It showed a 94% decrease in error rates during task execution.
- Integrated risk checks and continuous monitoring mechanisms improved compliance.

To successfully build and deploy Agentic AI from concept to reality, enterprises need to balance technical design with operational governance. This checklist walks you through each phase to break down goals and launch Agentic AI.

1. Hierarchical Planning: Break complex tasks into smaller goals and checkpoints.

Customization: Based on the complexity of a task and your risk appetite, [set the enterprise's goals](#) and choose how often to review the task's progress.

2. Memory Integration: Use short- and long-term memory to enable continuity and prevent hallucinations.

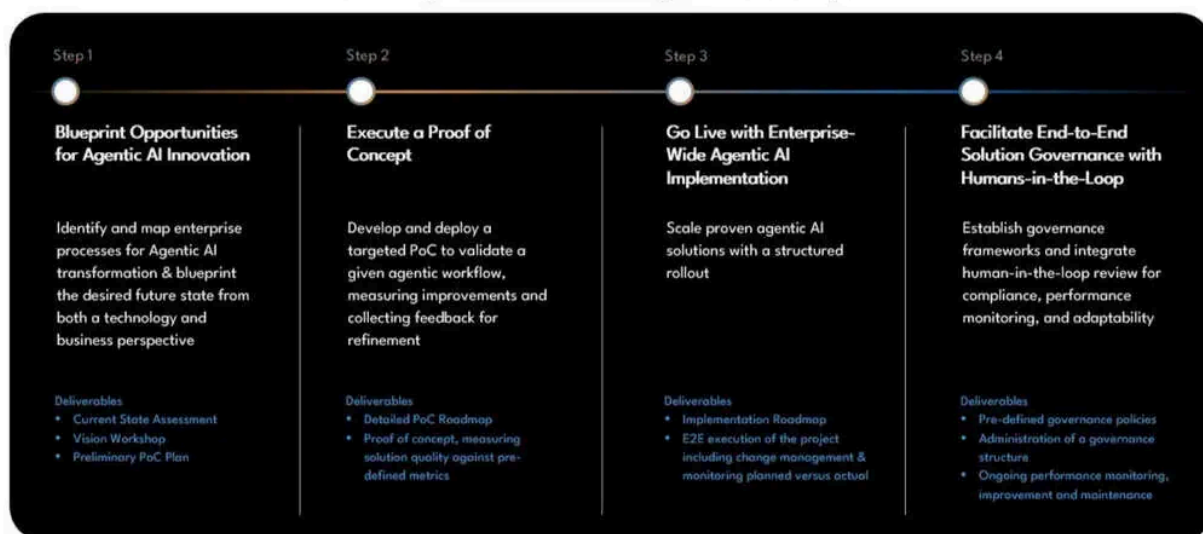
Customization: Adjust memory windows and integrate with your proprietary data sources like CRMs and document archives to reflect your domain knowledge.

3. Tool Orchestration Optimization: Enable the AI to pick and switch tools or APIs dynamically, optimizing workflows in real time.

Customization: Set performance thresholds and fallback options to ensure every task is completed reliably and efficiently.

How Innodata Can Help Facilitate Agentic Workflows

Our 4-Step Process to Promote Agentic AI in Enterprises



4. Observability & Monitoring: Embed real-time telemetry and [red teaming](#) to capture inputs, decisions, outputs, and performance metrics. This builds trust and accountability.

5. Human-in-the-loop Governance: Design checkpoints, fallback plans, and observability to ensure oversight and alignment.

Customization: Configure approval workflows, audit-log detail levels, and when to escalate problems to comply with your governance framework.

What Enterprises Often Overlook When Building Custom Agentic AI Architectures

While designing Agentic AI systems, organizations often underestimate some critical integration, operational, and ethical challenges. Addressing these overlooked factors leads to ensuring trust, compliance, and lasting business impact.

- **Tool Integration Complexity:** Connecting diverse tools and legacy systems can create silos and slow progress. Modular design and clear APIs can help simplify integration, support scaling, and cut maintenance costs.
- **Error Handling:** Complex, multi-step decision processes introduce potential failure points that can disrupt operations. Embedding retry logic, circuit breakers, and fallback workflows keeps systems resilient and prevents such shortfalls.
- **Latency Management:** Meeting real-time decision requirements without compromising performance is often underestimated. Optimizing data pipelines and parallelizing tasks helps maintain accuracy and responsiveness.
- **Ethical & Regulatory Risks:** Enterprises sometimes favor speed and automation and end up overlooking fairness, transparency, and compliance. By integrating bias detection, explainability layers, privacy safeguards, and traceability, organizations can meet strict governance standards.
- **Risk of Unsupervised Decision-making:** Fully autonomous decisions can introduce hidden risks if left unchecked. Embedding [human-in-the-loop](#) checkpoints, escalation paths, and audit trails ensures critical actions remain [under control and verifiable](#).

Build Reliable Agentic AI with Innodata

Ready to take the next step? [Connect](#) with Innodata to design your [custom Agentic AI](#) architecture today!

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